

A regularized Newton-type method for box-constrained order-value optimization*

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August 4, 2026

Abstract

The order-value function of order p returns, at each point, the p th smallest value among a finite family of functions $f_1, \dots, f_m$; minimizing it recovers the minimization of the minimum and of the maximum of the family in the extreme cases $p = 1$ and $p = m$, and provides a flexible tool for robust estimation, in which a prescribed number of outlying observations are discarded. We propose a regularized method for minimizing the order-value function subject to box constraints. At each iteration the method minimizes a strongly convex model that combines a piecewise-linear term built from the active gradients, a curvature term in which each active component carries its own symmetric positive semidefinite matrix of bounded norm, and a quadratic regularization. The curvature matrices are otherwise arbitrary, so that quasi-Newton updates, Gauss–Newton matrices and modified exact Hessians are all admissible, and a purely first-order regularized method is recovered as the special case in which they vanish. We introduce an approximate optimality condition for the problem and prove that the method is well defined and attains the same worst-case iteration- and evaluation-complexity bounds as its first-order counterpart, so that the incorporation of curvature preserves the theoretical guarantees at no additional order of complexity. The applicability of the method to parameter-estimation problems with outliers is illustrated, and the practical effect of the curvature is examined empirically.

Keywords: Order-value optimization, regularized models, Newton-type methods, complexity, algorithms, applications.

Mathematics Subject Classification: 90C30, 65K05, 49M37, 90C60, 68Q25.

*This work has been funded by the Vice-Rectorate for Research, Outreach and Social Projection of the University of the Atlantic through the 2022 Internal Call for Proposals to Strengthen Research Groups with Tools for Creation, Innovation, and Research.

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1 Introduction

Many optimization problems involve objective functions that are obtained not by aggregating a collection of simpler functions, but by selecting one of them according to its rank. The prototypical example is the order-value function of order p , introduced in [1, 2]: given m functions $f_1, \dots, f_m$, its value at a point x is the p th smallest among $f_1(x), \dots, f_m(x)$. Functions of this kind belong to the broader class of generalized order-value functions—those whose value at x is governed by order relations among the elements of a set $\{f_i(x)\}_{i \in I}$ —a class systematized in [14]. Since the index that attains the p th position may switch abruptly as x moves, the order-value function is generally nonsmooth even when each f_i is continuously differentiable.

The extreme choices $p = 1$ and $p = m$ recover the minimum and the maximum of $\{f_i(x)\}$, respectively, so the order-value function interpolates between these two classical aggregates; the intermediate cases are the ones of practical interest. When x encodes portfolio positions and $f_i(x)$ is the loss incurred under scenario i , the order-value function coincides with the discrete value-at-risk (VaR) measure, a standard risk measure in financial risk management [13]. When $f_i(x)$ measures the misfit of a parametric model to the i th observation, minimizing the p th order-value function fits the parameters while discarding the $m - p$ worst-fitted observations. This makes order-value optimization a natural tool for robust estimation, where one seeks to remain insensitive to a minority of grossly corrupted data, or outliers.

The order-value optimization (OVO) problem was first formulated in [1], where a steepest-descent-type method was proposed and shown to converge to points satisfying a weak optimality condition. Sharper optimality conditions and a reformulation as a nonlinear program with equilibrium constraints were subsequently developed in [2]. A quasi-Newton method generalizing the Cauchy scheme of [1]—approximating the active components by convex quadratics and proving global and local convergence—was introduced in [3], and a global optimization strategy combining multistart with a tunneling technique was proposed in [5]. A closely related variant, in which the sum of the p smallest values is minimized—low order-value optimization (LOVO)—was developed with applications to robust estimation and data analysis in [4]. On the computational-complexity side, [17] established that minimizing the order-value function over a polytope is NP-hard in the strong sense.

The systematic study of worst-case complexity in continuous optimization originates in the cubic-regularization analysis of [16], and complexity guarantees have since been obtained for a broad range of problems (see, e.g., [7, 8] and the monograph [11]). Within this line, [18] recently proposed a first-order method for box-constrained OVO that minimizes, at each iteration, a quadratically regularized piecewise-linear model of f , and established iteration- and evaluation-complexity bounds for reaching an approximate optimality condition.

In the present work we generalize the approach of [18] by enriching the regularized model with second-order information. At each iteration our method minimizes a model built from the same piecewise-linear convex term, a curvature term in which each active component f_i carries its own symmetric positive semidefinite matrix B_i of norm at most M , and a Tikhonov-type quadratic regularization. Beyond these two requirements the matrices B_i are arbitrary: a quasi-Newton update such as BFGS or SR1, a Gauss–Newton matrix, or an exact component Hessian shifted to positive definiteness are all admissible, and the analysis is indifferent to which is used. The regularization renders the model strongly convex, so that each subproblem has a

unique minimizer. Setting $B_i = 0$ recovers the first-order method of [18] exactly, so the proposed method is a strict generalization of it. Curvature had already been incorporated into OVO by the quasi-Newton method of [3], which approximates the active components by convex quadratics with positive semidefinite quasi-Newton Hessians and establishes global convergence together with local superlinear and quadratic rates; that method, however, controls the step through a trust-region bound combined with a line search, and its guarantees are asymptotic. Here the curvature is instead embedded in the adaptively regularized model of [18]: the Tikhonov term renders each subproblem strongly convex with a unique minimizer, and the analysis yields non-asymptotic worst-case iteration- and evaluation-complexity bounds rather than asymptotic convergence rates. We derive an approximate optimality condition for the box-constrained problem and prove that the method is well defined and attains the *same* worst-case iteration- and evaluation-complexity bounds as the first-order method (Theorem 3.9). Thus the incorporation of curvature does not degrade the theoretical guarantees; whether it also yields a practical advantage is a separate question, which we examine empirically on parameter-estimation problems in the presence of outliers.

The remainder of the paper is organized as follows. Section 2 defines the box-constrained order-value optimization problem and the proposed regularized method. Section 3 introduces an appropriate optimality condition and establishes the well-definiteness, convergence, and complexity of the method. Section 4 reports illustrative numerical experiments on parameter estimation in the presence of outliers. Final remarks are given in Section 5.

Notation. The symbol $\|\cdot\|$ denotes the Euclidean norm. For $i = 1, \dots, n$, $e_i \in \mathbb{R}^n$ denotes the i th column of the identity matrix.

2 A regularized Newton-type method

Let $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1, \dots, m$, be given and let $p \in \{1, \dots, m\}$ be fixed. The *order-value function of order p* is the mapping $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$f(x) = f_{i_p(x)}(x), \quad (1)$$

where $\{i_1(x), \dots, i_m(x)\}$ is a permutation of $\{1, \dots, m\}$ such that

$$f_{i_1(x)}(x) \leq f_{i_2(x)}(x) \leq \dots \leq f_{i_m(x)}(x); \quad (2)$$

that is, $f(x)$ is the p th smallest value among $f_1(x), \dots, f_m(x)$. The boundary cases $p = 1$ and $p = m$ recover, respectively, the minimum and the maximum of $\{f_1(x), \dots, f_m(x)\}$.

If $f_1, \dots, f_m$ are continuous, then so is f . Differentiability, however, is not inherited: f may be nonsmooth at a point x at which the p th smallest value is attained by more than one index, i.e. whenever there exist $i \neq j$ with $f_i(x) = f_j(x) = f(x)$, since the active index $i_p(x)$ may then change as x varies.

We address the box-constrained order-value optimization problem

$$\underset{x \in \Omega}{\text{Minimize}} \ f(x), \quad (3)$$

105 where $\Omega := \{x \in \mathbb{R}^n \mid \ell \leq x \leq u\}$ with $\ell, u \in \mathbb{R}^n$ and $\ell_i < u_i$ for $i = 1, \dots, n$. Box con-
 106 straints arise naturally in parameter estimation, where the unknowns are confined to physically
 107 or statistically meaningful bounds.

108 For $\delta > 0$ and $x \in \Omega$, define the *active index set*

$$109 \quad I_\delta(x) := \{i \in \{1, \dots, m\} \mid f(x) - \delta \leq f_i(x) \leq f(x) + \delta\}, \quad (4)$$

110 the set of indices whose values lie within a band of half-width δ around $f(x)$. As $\delta \rightarrow 0$, $I_\delta(x)$
 111 shrinks to the set of indices that attain the order-value,

$$112 \quad I_0(x) := \{i \in \{1, \dots, m\} \mid f_i(x) = f(x)\}.$$

113 The method builds, around a reference point $\bar{x} \in \Omega$, a local model of f that combines the
 114 first-order information of the active components with curvature. Each component carries its
 115 own curvature matrix: given $M > 0$, we call

$$116 \quad \mathcal{B} = \{B_i\}_{i=1}^m, \quad B_i \in \mathbb{R}^{n \times n} \text{ symmetric, } B_i \succeq 0, \quad \|B_i\| \leq M,$$

117 an *admissible curvature family*; only those B_i with $i \in I_\delta(\bar{x})$ enter the model below. For given
 118 $\bar{x} \in \Omega$, an admissible $\mathcal{B}$, $\delta \geq 0$, and $\sigma > 0$, the regularized model is

$$119 \quad \Psi(x; \bar{x}, \mathcal{B}, \delta, \sigma) = \max_{i \in I_\delta(\bar{x})} \left\{ \nabla f_i(\bar{x})^T (x - \bar{x}) + \frac{1}{2} [(x - \bar{x})^T B_i (x - \bar{x}) + \sigma \|x - \bar{x}\|^2] \right\}. \quad (5)$$

120 The first term is the piecewise-linear convex function that captures the worst-case first-order
 121 behavior of f over the active components; the bracket adds the curvature $(x - \bar{x})^T B_i (x - \bar{x})$ of the
 122 i th component and a Tikhonov regularization $\sigma \|x - \bar{x}\|^2$. Each component of the maximum is a
 123 convex quadratic, so Ψ is convex; since $\sigma > 0$ it is in fact strongly convex even when the B_i are
 124 only positive semidefinite, and the subproblem (6) has a unique minimizer. Taking $B_i = 0$ for all
 125 i reduces the model to the first-order regularized model of [18], which the present method thus
 126 generalizes; taking a single matrix $B_i \equiv B$ common to all active components is the intermediate
 127 case in which one curvature matrix is shared.

128 **Algorithm 2.1.** Let $\delta > 0$, $\sigma_{\min} > 0$, $M > 0$, $\alpha \in (0, 1)$, $\gamma > 1$, and $x^0 \in \Omega$ be given. Initialize
 129 $k \leftarrow 0$.

130 **Step 1.** Initialize $j \leftarrow 0$ and $\sigma_{k,j} = \sigma_{\min}$.

131 **Step 2.** Choose an admissible curvature family $\mathcal{B}_{k,j} = \{B_{k,j,i}\}_{i=1}^m$ and compute $x_{\text{trial}}^{k,j}$ as the
 132 global minimizer of

$$133 \quad \underset{x \in \Omega}{\text{Minimize}} \quad \Psi(x; x^k, \mathcal{B}_{k,j}, \delta, \sigma_{k,j}). \quad (6)$$

134 **Step 3.** If

$$135 \quad f(x_{\text{trial}}^{k,j}) \leq f(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2 \quad (7)$$

136 does not hold, set $\sigma_{k,j+1} = \gamma \sigma_{k,j}$, $j \leftarrow j + 1$, and go to Step 2.

137 **Step 4.** Set $x^{k+1} = x_{\text{trial}}^{k,j}$, $\sigma_k = \sigma_{k,j}$, $j_k = j$, $k \leftarrow k + 1$, and go to Step 1.

The bound $\|B_{k,j,i}\| \leq M$ keeps the curvature from dominating the model. Positive semidefiniteness and that bound are the *only* requirements: the freedom in choosing the $B_{k,j,i}$ lets the method accommodate standard quasi-Newton updates such as BFGS or SR1, Gauss–Newton matrices, and exact component Hessians shifted to positive definiteness, while the choice $B_{k,j,i} = 0$ for all k, j, i recovers the first-order method of [18]. The analysis below is indifferent to which of these is used.

The requirements on $x_{\text{trial}}^{k,j}$ are mild. The theory only needs $x_{\text{trial}}^{k,j}$ to be a stationary point of (6) with $\Psi(x_{\text{trial}}^{k,j}; x^k, \mathcal{B}_{k,j}, \delta, \sigma_{k,j}) \leq 0$; since $\Psi(x^k; x^k, \mathcal{B}_{k,j}, \delta, \sigma_{k,j}) = 0$, any descent method started at x^k produces such iterates. Moreover, (6) minimizes a strongly convex function over the compact convex set Ω , so it has a unique global minimizer and every stationary point is global; the global-minimizer requirement of Step 2 is therefore met by any such descent method, and standard convex solvers apply directly.

3 Optimality condition and complexity

Assumption A1. *The functions $f_1, \dots, f_m$ belong to $\mathcal{C}^1(\Omega)$; that is, they are continuously differentiable on Ω .*

We first show that a local minimizer of (3) also minimizes the model Ψ built at the corresponding reference point. These results connect the original nonsmooth problem with the convex subproblems solved at each iteration and underlie the optimality condition introduced below.

Theorem 3.1. *Suppose that Assumption A1 holds. If x^* is a local minimizer of (3), then it is also a local minimizer of $\text{Minimize}_{x \in \Omega} \Psi(x; x^*, 0, 0, 0)$.*

Proof. Suppose not. Then there is $y^* \in \Omega$ with $\Psi(y^*; x^*, 0, 0, 0) < \Psi(x^*; x^*, 0, 0, 0) = 0$, i.e. $\nabla f_i(x^*)^T(y^* - x^*) < 0$ for all $i \in I_0(x^*)$. By Assumption A1, for each such i ,

$$\lim_{t \rightarrow 0^+} \frac{f_i(x^* + t(y^* - x^*)) - f_i(x^*)}{t} = \nabla f_i(x^*)^T(y^* - x^*) < 0,$$

so there is $\bar{t}_i > 0$ with $f_i(x^* + t(y^* - x^*)) < f_i(x^*) = f(x^*)$ for $t \in (0, \bar{t}_i]$. With $\bar{t} = \min_{i \in I_0(x^*)} \bar{t}_i$,

$$f_i(x^* + t(y^* - x^*)) < f(x^*) \quad \text{for all } i \in I_0(x^*), \quad t \in (0, \bar{t}]. \quad (8)$$

For $j \notin I_0(x^*)$, continuity of $f_1, \dots, f_m$ gives, for t small,

$$f_i(x^* + t(y^* - x^*)) < f_j(x^* + t(y^* - x^*)) \quad \text{if } i \in I_0(x^*), \quad f(x^*) < f_j(x^*), \quad (9)$$

$$f_i(x^* + t(y^* - x^*)) > f_j(x^* + t(y^* - x^*)) \quad \text{if } i \in I_0(x^*), \quad f(x^*) > f_j(x^*). \quad (10)$$

Hence the ranking positions of the active indices $I_0(x^*)$ within $\{f_i(x^* + t(y^* - x^*))\}_{i=1}^m$ are preserved for t small, so the p th position is still occupied by some $i \in I_0(x^*)$ and $f(x^* + t(y^* - x^*)) = f_i(x^* + t(y^* - x^*))$. With (8), $f(x^* + t(y^* - x^*)) < f(x^*)$ for t small. Since Ω is convex, $x^* + t(y^* - x^*) \in \Omega$, contradicting the local minimality of x^* . $\square$

171 **Theorem 3.2.** Suppose that Assumption A1 holds and let $\mathcal{B}$ be an admissible curvature family
 172 and $\sigma \geq 0$. If x^* is a local minimizer of (3), then it is also a local minimizer of

$$173 \quad \underset{x \in \Omega}{\text{Minimize}} \Psi(x; x^*, \mathcal{B}, 0, \sigma).$$

174 *Proof.* Suppose not: there is $y^* \in \Omega$ with $\Psi(y^*; x^*, \mathcal{B}, 0, \sigma) < 0$. Since the model is a maximum
 175 over $i \in I_0(x^*)$, every one of its components is negative at y^* :

$$176 \quad \nabla f_i(x^*)^T (y^* - x^*) + \frac{1}{2} [(y^* - x^*)^T B_i (y^* - x^*) + \sigma \|y^* - x^*\|^2] < 0 \quad \text{for all } i \in I_0(x^*).$$

177 Since $B_i \succeq 0$ and $\sigma \geq 0$, each bracket is nonnegative, so $\nabla f_i(x^*)^T (y^* - x^*) < 0$ for every
 178 $i \in I_0(x^*)$, and therefore $\max_{i \in I_0(x^*)} \{\nabla f_i(x^*)^T (y^* - x^*)\} < 0$, that is, $\Psi(y^*; x^*, \mathcal{O}, 0, 0) < 0$,
 179 where $\mathcal{O}$ denotes the admissible family with $B_i = 0$ for all i . This contradicts Theorem 3.1. $\square$

180 **Theorem 3.3.** Suppose that Assumption A1 holds and let $\mathcal{B}$ be an admissible curvature fam-
 181 ily and $\sigma, \delta \geq 0$. If x^* is a local minimizer of (3), then it is also a local minimizer of
 182 $\underset{x \in \Omega}{\text{Minimize}} \Psi(x; x^*, \mathcal{B}, \delta, \sigma)$.

183 *Proof.* Suppose not: there is $y^* \in \Omega$ with $\Psi(y^*; x^*, \mathcal{B}, \delta, \sigma) < 0$. Both models are maxima of the
 184 same family of functions, taken over $I_0(x^*)$ and over $I_\delta(x^*)$ respectively; since $I_0(x^*) \subseteq I_\delta(x^*)$,
 185 the maximum over the smaller index set is no larger, so

$$186 \quad \Psi(y^*; x^*, \mathcal{B}, 0, \sigma) \leq \Psi(y^*; x^*, \mathcal{B}, \delta, \sigma) < 0,$$

187 contradicting Theorem 3.2. $\square$

188 Throughout, $\Sigma_q := \{\lambda \in \mathbb{R}^q \mid \sum_{j=1}^q \lambda_j = 1, \lambda_j \geq 0\}$ denotes the unit simplex. Following
 189 [18], the next consequence of Theorem 3.3 provides a first-order necessary condition for (3) and
 190 motivates the stopping criterion below.

191 **Corollary 3.4.** Suppose that Assumption A1 holds and let $\delta \geq 0$. If x^* is a local minimizer of
 192 (3), then there exist $\mu \in \Sigma_{|I_\delta(x^*)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}_+^n$ such that $(\ell_i - x_i^*)\nu_i^\ell = (x_i^* - u_i)\nu_i^u = 0$ for
 193 $i = 1, \dots, n$ and

$$194 \quad \sum_{i \in I_\delta(x^*)} \mu_i \nabla f_i(x^*) + \sum_{i=1}^n (\nu_i^u - \nu_i^\ell) e_i = 0. \quad (11)$$

195 *Proof.* By Theorem 3.3 with $B_i = 0$ for all i and $\sigma = 0$, the point x^* minimizes $x \mapsto$
 196 $\max_{i \in I_\delta(x^*)} \{\nabla f_i(x^*)^T (x - x^*)\}$ over Ω . Applying the KKT conditions for the minimization
 197 of a maximum of differentiable functions over a box (Lemma 3.1 of [18]) yields multipliers
 198 $\mu \in \Sigma_{|I_\delta(x^*)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}_+^n$ with the stated complementarity and (11). $\square$

199 **Definition 3.5.** Given $\epsilon > 0$ and $\delta > 0$, we say that $x \in \Omega$ satisfies the approximate optimality
 200 condition $C_{\delta, \epsilon}$ if there exist $\mu \in \Sigma_{|I_\delta(x)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}_+^n$ such that $(\ell_i - x_i)\nu_i^\ell = (x_i - u_i)\nu_i^u = 0$
 201 for $i = 1, \dots, n$ and

$$202 \quad \left\| \sum_{i \in I_\delta(x)} \mu_i \nabla f_i(x) + \sum_{i=1}^n (\nu_i^u - \nu_i^\ell) e_i \right\| \leq \epsilon. \quad (12)$$

Condition $C_{\delta,\epsilon}$ requires the gradient combination of the active components, corrected by the box multipliers, to be small in norm. For $\epsilon = 0$ it reduces to the exact condition (11) Corollary 3.4. It serves as the stopping criterion for Algorithm 2.1.

Corollary 3.6. *Suppose that Assumption A1 holds. Then, for every k and $j = 0, \dots, j_k$, there exist $\mu \in \Sigma_{|I_\delta(x^k)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}_+^n$ such that $(\ell_i - [x_{\text{trial}}^{k,j}]_i)\nu_i^\ell = ([x_{\text{trial}}^{k,j}]_i - u_i)\nu_i^u = 0$ for $i = 1, \dots, n$ and*

$$x_{\text{trial}}^{k,j} - x^k = (\bar{B}_{k,j} + \sigma_{k,j}I)^{-1} \left[\sum_{i=1}^n (\nu_i^\ell - \nu_i^u) e_i - \sum_{i \in I_\delta(x^k)} \mu_i \nabla f_i(x^k) \right], \quad (13)$$

where

$$\bar{B}_{k,j} := \sum_{i \in I_\delta(x^k)} \mu_i B_{k,j,i} \quad (14)$$

is symmetric, positive semidefinite, and satisfies $\|\bar{B}_{k,j}\| \leq M$.

Proof. Subproblem (6) minimizes the strongly convex function $\Psi(\cdot; x^k, \mathcal{B}_{k,j}, \delta, \sigma_{k,j})$ over the box Ω . By the KKT conditions for a maximum of differentiable functions over a box (Lemma 3.1 of [18]), there exist $\mu \in \Sigma_{|I_\delta(x^k)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}_+^n$ with the stated complementarity such that

$$\sum_{i \in I_\delta(x^k)} \mu_i [\nabla f_i(x^k) + (B_{k,j,i} + \sigma_{k,j}I)(x_{\text{trial}}^{k,j} - x^k)] + \sum_{i=1}^n (\nu_i^u - \nu_i^\ell) e_i = 0,$$

where $\nabla f_i(x^k) + (B_{k,j,i} + \sigma_{k,j}I)(x_{\text{trial}}^{k,j} - x^k)$ is the gradient at $x_{\text{trial}}^{k,j}$ of the i th component of Ψ . Since $\sum_i \mu_i = 1$, the terms involving $\sigma_{k,j}$ collect into $\sigma_{k,j}(x_{\text{trial}}^{k,j} - x^k)$ and those involving the curvature into $\bar{B}_{k,j}(x_{\text{trial}}^{k,j} - x^k)$ with $\bar{B}_{k,j}$ given by (14). Being a convex combination of symmetric positive semidefinite matrices, $\bar{B}_{k,j}$ is itself symmetric and positive semidefinite, and $\|\bar{B}_{k,j}\| \leq \sum_{i \in I_\delta(x^k)} \mu_i \|B_{k,j,i}\| \leq M$. As $\sigma_{k,j} > 0$ and $\bar{B}_{k,j} \succeq 0$, the matrix $\bar{B}_{k,j} + \sigma_{k,j}I$ is positive definite, and solving for $x_{\text{trial}}^{k,j} - x^k$ gives (13). $\square$

Assumption A2. *For all k and $j = 0, \dots, j_k$, the multipliers ν^ℓ, ν^u of Corollary 3.6 satisfy $\max_{i=1, \dots, n} |\nu_i^\ell - \nu_i^u| \leq c_\nu$ for some constant $c_\nu > 0$ independent of k and j .*

The Mangasarian–Fromovitz constraint qualification guarantees that, for each (k, j) , these multipliers are bounded [12]; Assumption A2 strengthens this to a single bound uniform in k and j , which is natural since Ω is compact and the active gradients are bounded under Assumption A1.

Assumption A3. *There is a constant $c_\nabla > 0$ such that $\|\nabla f_i(x)\| \leq c_\nabla$ for all $i = 1, \dots, m$ and all $x \in \Omega$.*

Assumption A3 holds automatically under Assumption A1, since Ω is bounded. Combining Corollary 3.6 with Assumptions A2 and A3 bounds the step. Since $\bar{B}_{k,j} \succeq 0$, the eigenvalues of $\bar{B}_{k,j} + \sigma_{k,j}I$ are at least $\sigma_{k,j}$, so $\|(\bar{B}_{k,j} + \sigma_{k,j}I)^{-1}\| \leq 1/\sigma_{k,j}$, and (13) gives

$$\|x_{\text{trial}}^{k,j} - x^k\| \leq \frac{1}{\sigma_{k,j}} \left(\sum_{i=1}^n |\nu_i^\ell - \nu_i^u| + \sum_{i \in I_\delta(x^k)} \mu_i \|\nabla f_i(x^k)\| \right) \leq \frac{n c_\nu + c_\nabla}{\sigma_{k,j}},$$

235 where the last inequality uses Assumptions A2, A3 and $\mu \in \Sigma_{|I_\delta(x^k)|}$. Thus

$$236 \quad \|x_{\text{trial}}^{k,j} - x^k\| \leq \frac{c_x}{\sigma_{k,j}}, \quad c_x := n c_\nu + c_\nabla. \quad (15)$$

237 The step shrinks as $\sigma_{k,j}$ grows, which is the mechanism driving the acceptance of (7).

238 **Assumption A4.** *There is $L > 0$ such that $\|\nabla f_i(y) - \nabla f_i(x)\| \leq L\|y - x\|$ for all $i = 1, \dots, m$*
 239 *and $x, y \in \Omega$.*

240 Assumption A4 is the standard gradient-Lipschitz condition. It implies, for all i and $x, y \in \Omega$,

$$241 \quad f_i(y) \leq f_i(x) + \nabla f_i(x)^T(y - x) + \frac{L}{2}\|y - x\|^2, \quad (16)$$

242 as well as the two-sided bound $|f_i(y) - f_i(x) - \nabla f_i(x)^T(y - x)| \leq \frac{L}{2}\|y - x\|^2$. These make Ψ
 243 an upper model of f once $\sigma_{k,j}$ is large relative to L , which yields finite termination of the inner
 244 loop.

245 **Lemma 3.7.** *Let $a_1, \dots, a_r, b_1, \dots, b_r \in \mathbb{R}$ and $\beta > 0$ satisfy $a_1 \leq \dots \leq a_r$, $b_j \leq a_j - \beta$ for*
 246 *$j = 1, \dots, r$, and $b_{i_1} \leq \dots \leq b_{i_r}$ for a permutation $\{i_1, \dots, i_r\}$ of $\{1, \dots, r\}$. Then $b_{i_q} \leq a_q - \beta$*
 247 *for $q = 1, \dots, r$.*

248 *Proof.* See Lemma 2.1 of [1]. □

249 The next result gives a threshold on $\sigma_{k,j}$ beyond which the trial point is accepted at Step 3,
 250 so that the inner loop terminates after finitely many increases of $\sigma_{k,j}$, independently of k . The
 251 threshold scales with δ^{-1} : a narrower active band demands stronger regularization.

252 **Theorem 3.8.** *Suppose that Assumptions A1, A3, and A4 hold. Then, for all k and $j =$*
 253 *$0, \dots, j_k$, if*

$$254 \quad \sigma_{k,j} \geq \max\{L + 2\alpha, 9c_x^2/(2\delta)\}, \quad (17)$$

255 *then $x_{\text{trial}}^{k,j}$ satisfies (7).*

256 *Proof.* Assume (17); in particular $\sigma_{k,j} \geq L + 2\alpha$. By (16), for every $i \in \{1, \dots, m\}$,

$$\begin{aligned} 257 \quad f_i(x_{\text{trial}}^{k,j}) &\leq f_i(x^k) + \nabla f_i(x^k)^T(x_{\text{trial}}^{k,j} - x^k) + \frac{L}{2}\|x_{\text{trial}}^{k,j} - x^k\|^2 \\ 258 \quad &\leq f_i(x^k) + \nabla f_i(x^k)^T(x_{\text{trial}}^{k,j} - x^k) + \left(\frac{\sigma_{k,j}}{2} - \alpha\right)\|x_{\text{trial}}^{k,j} - x^k\|^2 \\ 259 \quad &= f_i(x^k) + \nabla f_i(x^k)^T(x_{\text{trial}}^{k,j} - x^k) \\ 260 \quad &\quad + \frac{1}{2}[(x_{\text{trial}}^{k,j} - x^k)^T B_{k,j,i}(x_{\text{trial}}^{k,j} - x^k) + \sigma_{k,j}\|x_{\text{trial}}^{k,j} - x^k\|^2] \\ 261 \quad &\quad - \alpha\|x_{\text{trial}}^{k,j} - x^k\|^2 - \frac{1}{2}(x_{\text{trial}}^{k,j} - x^k)^T B_{k,j,i}(x_{\text{trial}}^{k,j} - x^k). \end{aligned}$$

262 Since $B_{k,j,i} \succeq 0$, the last term is nonpositive and may be dropped. For $i \in I_\delta(x^k)$, the mini-
 263 mizer property $\Psi(x_{\text{trial}}^{k,j}; x^k, \mathcal{B}_{k,j}, \delta, \sigma_{k,j}) \leq \Psi(x^k; x^k, \mathcal{B}_{k,j}, \delta, \sigma_{k,j}) = 0$ gives, since the model is a
 264 maximum over $I_\delta(x^k)$ and therefore bounds each of its components,

$$265 \quad \nabla f_i(x^k)^T(x_{\text{trial}}^{k,j} - x^k) + \frac{1}{2}[(x_{\text{trial}}^{k,j} - x^k)^T B_{k,j,i}(x_{\text{trial}}^{k,j} - x^k) + \sigma_{k,j}\|x_{\text{trial}}^{k,j} - x^k\|^2] \leq 0,$$

266 so that

$$267 \quad f_i(x_{\text{trial}}^{k,j}) \leq f_i(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2, \quad i \in I_\delta(x^k). \quad (18)$$

268 On the other hand, by (15), the Cauchy–Schwarz inequality, and the two-sided consequence of
 269 Assumption A4, using $\|\nabla f_i(x^k)\| \leq c_\nabla \leq c_x$ and (15), $|f_i(x_{\text{trial}}^{k,j}) - f_i(x^k)| \leq c_x^2/\sigma_{k,j} + Lc_x^2/(2\sigma_{k,j}^2)$
 270 for all i ; using $L \leq \sigma_{k,j}$, this is at most $3c_x^2/(2\sigma_{k,j})$, and (17) yields

$$271 \quad |f_i(x_{\text{trial}}^{k,j}) - f_i(x^k)| \leq \frac{\delta}{3}, \quad i = 1, \dots, m. \quad (19)$$

272 Consequently, any index with $f_i(x^k) < f(x^k) - \delta$ satisfies $f_i(x_{\text{trial}}^{k,j}) < f(x^k)$, and any index with
 273 $f_i(x^k) > f(x^k) + \delta$ satisfies $f_i(x_{\text{trial}}^{k,j}) > f(x^k)$. In particular $i_p(x_{\text{trial}}^{k,j}) \in I_\delta(x^k)$, and the number of
 274 components lying below the band is the same at x^k and $x_{\text{trial}}^{k,j}$. Write $I_\delta(x^k) = \{i_1, \dots, i_r\}$ with
 275 $f_{i_1}(x^k) \leq \dots \leq f_{i_r}(x^k)$, and let $\{i'_1, \dots, i'_r\}$ reorder them so that $f_{i'_1}(x_{\text{trial}}^{k,j}) \leq \dots \leq f_{i'_r}(x_{\text{trial}}^{k,j})$.
 276 Since the count below the band is preserved and $i_p(\cdot)$ lies in the band at both points, $i_p(x^k)$ and
 277 $i_p(x_{\text{trial}}^{k,j})$ occupy the same position q within the band; that is, $i_p(x^k) = i_q$ and $i_p(x_{\text{trial}}^{k,j}) = i'_q$.
 278 Applying Lemma 3.7 with $a_j = f_{i_j}(x^k)$, $b_j = f_{i_j}(x_{\text{trial}}^{k,j})$, and $\beta = \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2$ —valid by
 279 (18)—gives $f_{i'_q}(x_{\text{trial}}^{k,j}) \leq f_{i_q}(x^k) - \beta$. Since $f_{i_q}(x^k) = f_{i_p(x^k)}(x^k) = f(x^k)$,

$$280 \quad f(x_{\text{trial}}^{k,j}) = f_{i_p(x_{\text{trial}}^{k,j})}(x_{\text{trial}}^{k,j}) = f_{i'_q}(x_{\text{trial}}^{k,j}) \leq f(x^k) - \alpha \|x_{\text{trial}}^{k,j} - x^k\|^2,$$

281 which is (7). □

282 The inner loop thus terminates finitely at every outer iteration. The curvature affects only
 283 the constants below—through the bound M on $\|B_{k,j,i}\|$ —and not the $O(\delta^{-2}\epsilon^{-2})$ iteration
 284 order, which coincides with that of the first-order method [18].

285 **Theorem 3.9.** *Suppose that Assumptions A1, A3, and A4 hold and that there is $f_{\text{low}} \in \mathbb{R}$ with*
 286 *$f(x) \geq f_{\text{low}}$ for all $x \in \Omega$. Then*

$$287 \quad \sigma_k \leq \gamma \max\{L + 2\alpha, 9c_x^2/(2\delta)\}, \quad (20)$$

288 and at most

$$289 \quad \lceil 1 + \log_\gamma(\sigma_k/\sigma_{\min}) \rceil \quad (21)$$

290 function evaluations are performed per outer iteration to obtain (7). Moreover, the number of
 291 outer iterations k at which $C_{\delta,\epsilon}$ is not satisfied by x^{k+1} is at most

$$292 \quad \left\lceil \frac{(M + \gamma \max\{L + 2\alpha, 9c_x^2/(2\delta)\})^2}{\alpha} \cdot \frac{f(x^0) - f_{\text{low}}}{\epsilon^2} \right\rceil. \quad (22)$$

293 *Proof.* Bounds (20) and (21) follow from Theorem 3.8 and the rule $\sigma_{k,j+1} = \gamma\sigma_{k,j}$: starting from
 294 $\sigma_{k,0} = \sigma_{\min}$, at most $\lceil 1 + \log_\gamma(\sigma_k/\sigma_{\min}) \rceil$ increases occur before (17) holds, after which (7) is
 295 satisfied.

Let K be the set of outer indices k at which $C_{\delta,\epsilon}$ fails at x^{k+1} . Fix $k \in K$ and write $\bar{B}_k = \bar{B}_{k,j_k}$, the convex combination (14) associated with the accepted trial point. By Corollary 3.6, $(\bar{B}_k + \sigma_k I)(x^{k+1} - x^k) = \sum_{i=1}^n (\nu_i^\ell - \nu_i^u) e_i - \sum_{i \in I_\delta(x^k)} \mu_i \nabla f_i(x^k)$, hence

$$\|(\bar{B}_k + \sigma_k I)(x^{k+1} - x^k)\| = \left\| \sum_{i \in I_\delta(x^k)} \mu_i \nabla f_i(x^k) + \sum_{i=1}^n (\nu_i^u - \nu_i^\ell) e_i \right\| > \epsilon,$$

the inequality holding because $C_{\delta,\epsilon}$ fails at x^{k+1} . Since $\|\bar{B}_k\| \leq M$ by Corollary 3.6,

$$\epsilon < \|(\bar{B}_k + \sigma_k I)(x^{k+1} - x^k)\| \leq (M + \sigma_k) \|x^{k+1} - x^k\|,$$

and with (20), $\|x^{k+1} - x^k\| > \epsilon / (M + \gamma \max\{L + 2\alpha, 9c_x^2/(2\delta)\})$. As x^{k+1} satisfies (7),

$$f(x^{k+1}) \leq f(x^k) - \alpha \|x^{k+1} - x^k\|^2 < f(x^k) - \frac{\alpha \epsilon^2}{(M + \gamma \max\{L + 2\alpha, 9c_x^2/(2\delta)\})^2}.$$

Summing over $k \in K$ and using $f(x) \geq f_{\text{low}}$ gives $f(x^0) - f_{\text{low}} > |K| \cdot \alpha \epsilon^2 / (M + \gamma \max\{L + 2\alpha, 9c_x^2/(2\delta)\})^2$, from which (22) follows. $\square$

4 Numerical experiments

We illustrate the method on parameter-estimation problems in which the data contain an unknown number of outliers, following the OVO methodology of [18]. Given a model $y(t, x)$ with parameters $x \in \Omega$ and data (t_i, y_i) , $i = 1, \dots, m$, we set

$$f_i(x) = \frac{1}{2} (y(t_i, x) - y_i)^2, \quad i = 1, \dots, m.$$

For a prescribed number o of outliers, solving (3) with $p = m - o$ fits the parameters to the p best-explained observations while discarding the o worst. Since the true number of outliers is unknown, we solve a sequence of OVO problems with increasing o ; a sharp drop in $f(x^*)$ identifies the number of outliers present.

Our aim is to assess the effect of the curvature. We therefore compare two instances of Algorithm 2.1 that differ *only* in the curvature family: the first-order method ($B_{k,j,i} = 0$ for all i , which coincides with [18]) and a second-order variant, in which each active component contributes its own matrix. For the least-squares functions above, $\nabla f_i(x) = r_i(x) \nabla r_i(x)$ and

$$\nabla^2 f_i(x) = \nabla r_i(x) \nabla r_i(x)^T + r_i(x) \nabla^2 r_i(x), \quad r_i(x) = y(t_i, x) - y_i.$$

We take $B_{k,j,i}$ to be the exact component Hessian $\nabla^2 f_i(x^k)$, made positive definite by the shift

$$B_{k,j,i} \leftarrow \nabla^2 f_i(x^k) + \max\{0, -\lambda_{\min}(\nabla^2 f_i(x^k)) + \tau\} I, \quad \tau = 10^{-8}, \quad (23)$$

and then rescaled by $M/\lambda_{\max}(B_{k,j,i})$ whenever $\lambda_{\max}(B_{k,j,i}) > M$, so that the requirements $B_{k,j,i} \succeq 0$ and $\|B_{k,j,i}\| \leq M$ of Step 2 hold. The eigenvalues are computed with the LAPACK routine DSYEV. For the model of Section 4.1, which is affine in the parameters, $\nabla^2 r_i = 0$ and

the component Hessian reduces to the constant Gauss–Newton matrix $\nabla r_i \nabla r_i^T$, already positive semidefinite, so the shift in (23) is inactive up to τ ; for the nonlinear model of Section 4.2 the term $r_i \nabla^2 r_i$ is present, the component Hessians are indefinite in general, and the shift is genuinely active. Quasi-Newton updates such as BFGS or SR1 are equally admissible in Step 2; we use exact Hessians here because $\nabla^2 r_i$ is available in closed form for both test problems, which isolates the effect of the curvature itself from that of any approximation to it. Like δ , the value of M is chosen by trial and error.

As in [18], subproblem (6) is reformulated as the smooth nonlinear program

$$\text{Minimize } w \quad \text{s.t.} \quad \nabla f_i(x^k)^T(x-x^k) + \frac{1}{2}[(x-x^k)^T B_{k,j,i}(x-x^k) + \sigma_{k,j}\|x-x^k\|^2] \leq w, \quad i \in I_\delta(x^k), \\ x \in \Omega, w$$

solved with Algencan [9, 10], whose multipliers furnish μ, ν^ℓ, ν^u for checking $C_{\delta,\epsilon}$. Throughout we use $\sigma_{\min} = 0.1$, $\alpha = 10^{-8}$, $\gamma = 5$, and $\epsilon = 10^{-4}$. Both methods are run from identical starting points with the same random seed, so that any difference is attributable to B .

The codes were implemented in Fortran 90. Tests were conducted on a computer with a 13th-generation Intel Core i7-13700KF processor (up to 5.4 GHz) and 32 GB 4800 MT/s DDR5 RAM, running Linux (Arch Linux, kernel 7.1.4). Code was compiled by the GNU Fortran compiler (version 16.1.1) with the `-O3` optimization directive enabled.

4.1 A model linear in the parameters

Following [1], we fit the third-degree polynomial $y(t, x) = x_1 + x_2 t + x_3 t^2 + x_4 t^3$, with $-10 \leq x_i \leq 10$, to a set of $m = 46$ observations containing 10 outliers, taken from [1]. The least-squares fit to the contaminated data is $\bar{x} \approx (6.46, 2.71, -7.54, 2.16)^T$. As the OVO problem has many non-global minimizers, for each $o \in \{0, \dots, 12\}$ we run the algorithm from 100 starting points $x_i^0 = \bar{x}_i + r |\bar{x}_i|$, with $r \sim U[-0.5, 0.5]$ (projected onto Ω), and report the best x^* . We take $\delta = 0.1$ and $M = 1$.

Table 1 reports, for each o and for both methods, the best value $f(x^*)$ over the 100 starting points, the iterations and evaluations of the single run that attained it, and the mean and median iterations and evaluations over all 100 runs. Both methods identify the 10 outliers through the sharp drop of $f(x^*)$ between $o = 9$ and $o = 10$, and return solutions of the same quality. The “best run” counts are a single sample that depends on the particular starting point, and are reported only to show at what cost that particular solution was reached; the fair cost comparison is the mean and median over all 100 runs, on which the two methods are essentially identical. Summing over o the per- o means reported in Table 1 gives 368 iterations and 1836 evaluations for the first-order method against 369 and 1858 for the second-order variant; the corresponding sums of the per- o medians (the median columns of the table) agree just as closely, 350.0 and 1760.5 against 352.5 and 1792.0. The two methods thus differ only within the run-to-run variability of the multistart procedure. This is expected: on this affine model the component Hessians are constant and already positive semidefinite, so the explicit curvature term is redundant and the matrices $B_{k,j,i}$ have no discernible effect.

The absence of any gain from curvature on this problem is expected, and instructive. The model $y(t, x)$ is affine in the parameters x , so each residual r_i is affine, ∇r_i is constant, and the component Hessians $\nabla^2 f_i = \nabla r_i \nabla r_i^T$ are constant and positive semidefinite: here the Gauss–Newton matrix *is* the exact Hessian, and the shift (23) does nothing. Each f_i is thus a convex

o	First-order ($B = 0$)								Second-order ($M = 1$)							
	best run		#it		#fcnt		best run		#it		#fcnt					
	$f(x^*)$	#it	#fcnt	mean	med	mean	med	$f(x^*)$	#it	#fcnt	mean	med	mean	med		
0	1.363e+01	20	120	31	30.0	151	149.5	1.362e+01	45	205	30	29.5	147	143.5		
1	1.145e+01	45	238	30	30.0	146	142.0	1.145e+01	20	77	31	30.0	155	152.5		
2	1.004e+01	22	114	31	28.5	155	152.5	1.005e+01	17	84	30	28.5	152	150.0		
3	9.535e+00	40	182	29	28.0	147	143.5	9.530e+00	15	84	28	27.0	143	137.0		
4	9.013e+00	64	341	29	27.5	140	132.5	9.014e+00	33	170	28	27.0	138	132.0		
5	8.455e+00	38	170	29	27.0	144	133.0	8.460e+00	32	158	30	29.0	149	143.0		
6	7.436e+00	38	199	28	24.0	143	126.0	7.434e+00	40	193	29	26.5	147	135.5		
7	6.921e+00	31	151	30	30.0	151	150.5	6.907e+00	19	89	31	31.0	161	155.0		
8	5.503e+00	39	158	28	27.0	139	139.0	5.501e+00	32	139	28	26.0	143	140.5		
9	4.176e+00	33	150	26	24.0	132	117.5	4.190e+00	55	282	25	23.5	126	119.5		
10	3.120e-02	29	170	26	23.5	130	122.0	4.077e-02	35	182	26	24.0	133	123.0		
11	3.452e-02	27	137	25	25.0	126	122.5	2.275e-02	21	89	26	24.5	129	124.5		
12	3.228e-02	23	142	26	25.5	132	130.0	3.421e-02	22	116	27	26.0	135	136.0		
Σ		449	2272	368	350.0	1836	1760.5		386	1868	369	352.5	1858	1792.0		

Table 1: Third-degree polynomial ($m = 46$, $\delta = 0.1$): first-order method ($B = 0$) versus the second-order variant ($M = 1$). For each o , the “best run” block reports the best $f(x^*)$ over 100 starting points together with the outer iterations (#it) and function evaluations (#fcnt) of the single run that attained it; the remaining columns give the mean and median #it and #fcnt over all 100 starting points. The Σ row sums each count over o .

quadratic, for which the adaptive regularization $\sigma_{k,j}$ of the first-order model already produces well-scaled steps; the explicit curvature is largely redundant and merely reshapes the step without reducing the iteration count. Moreover, the dominant difficulty of (3) on this instance is not smooth nonconvexity—which curvature would mitigate—but the combinatorial selection of which observations to discard, which produces the many local minima reflected in the multistart strategy and is unaffected by the curvature. This is consistent with the analysis of Section 3, where the curvature enters only the constant, and not the order, of the complexity bounds: on a problem of this type curvature can be expected to neither help nor harm. We turn next to a model that is genuinely nonlinear in the parameters, where the first-order model is a poorer local approximation, to examine whether the additional curvature translates into a practical advantage.

4.2 A model nonlinear in the parameters

Following [15], we fit the Bard function, whose i th residual is

$$r_i(x) = x_1 + \frac{u_i}{v_i x_2 + w_i x_3} - y_i, \quad u_i = t_i, \quad v_i = 16 - t_i, \quad w_i = \min(u_i, v_i), \quad (24)$$

where t_i is the abscissa of the i th observation, and set $f_i(x) = \frac{1}{2}r_i(x)^2$; the model has $n = 3$ parameters. The original data set of [15] consists of $m_0 = 15$ observations at $t_i = i$, $i = 1, \dots, 15$. The model is nonlinear in x_2 and x_3 , which enter through the denominator, so the component Hessians $\nabla^2 f_i = \nabla r_i \nabla r_i^T + r_i \nabla^2 r_i$ are no longer constant, the second term does not vanish, and the curvature matrices carry genuine point-dependent information; the shift (23) is active whenever $\nabla^2 f_i(x^k)$ is indefinite. We take the box $x_1 \in [-10, 10]$, $x_2, x_3 \in [0.1, 10]$; the lower

386 bounds on x_2, x_3 keep the denominator in (24) positive and exclude the spurious minimizer of
 387 the Bard function at $(0.8406\dots, -\infty, -\infty)$.

388 To the $m_0 = 15$ clean observations we add four outliers at random abscissas $t_i \in (2, 14)$,
 389 two lying above and two below the model curve, with random magnitudes clearly separated
 390 from the clean-data noise; drawn with a fixed random seed, they yield $m = 19$ observations.
 391 As in Section 4.1 we run the two variants from 100 starting points obtained by perturbing the
 392 standard initial guess $x^0 = (1, 1, 1)$, and report the best x^* ; we take $\delta = 0.1$ and $M = 1$. For
 393 each $o \in \{0, \dots, 6\}$ we solve (3) with $p = m - o$.

394 Table 2 reports the results. As the true number of outliers is unknown, one solves a sequence
 395 of problems with increasing o and looks for a sharp drop in $f(x^*)$; here $f(x^*)$ decreases only
 396 moderately for $o \leq 3$ and then drops by more than two orders of magnitude between $o = 3$ and
 397 $o = 4$, after which it remains negligible (Figure 1(b)). Both methods exhibit the same drop
 398 at $o = 4$ and both identify exactly the four added outliers. For $o \leq 3$ they return essentially
 399 identical values of $f(x^*)$; for $o \geq 4$, once the outliers are removed, both reduce $f(x^*)$ to the
 400 residual floor of the clean data, several orders of magnitude below the plateau, with the first-
 401 order method reaching the lower values there — by a factor of about 15 at $o = 6$ (6.3×10^{-5}
 402 against 9.5×10^{-4}). At that scale both variants have already discarded the four outliers and fit
 403 the clean data, and the remaining gap reflects which local minimizer the multistart happened
 404 to reach rather than a systematic difference in accuracy. Figure 1(a) shows the fit obtained at
 405 $o = 4$: the model follows the fifteen clean observations while the four outliers, marked with
 406 circles, are discarded.

o	First-order ($B = 0$)								Second-order ($M = 1$)							
	best run		#it		#fcnt		#it		best run		#it		#fcnt		#it	
	$f(x^*)$	#it	#fcnt	mean	med	mean	med	mean	med	$f(x^*)$	#it	#fcnt	mean	med	mean	med
0	2.643e+00	367	394	361	362.5	382	385.0	2.643e+00	393	411	399	397.5	416	414.5		
1	1.879e+00	8	30	152	10.5	166	29.5	1.876e+00	16	45	39	7.0	54	22.0		
2	6.765e-01	5	17	32	7.0	45	23.0	6.693e-01	4	12	38	7.0	49	18.0		
3	3.511e-01	7	21	13	8.0	28	22.0	3.549e-01	8	27	11	6.0	20	15.0		
4	2.086e-03	7	24	10	8.0	19	17.5	2.785e-03	2	5	8	6.0	13	11.0		
5	6.578e-05	6	27	16	5.0	22	11.0	1.021e-04	6	9	8	6.0	11	8.0		
6	6.319e-05	13	21	9	5.0	16	12.0	9.543e-04	7	8	10	6.0	13	9.0		
Σ		413	534	593	406.0	678	500.0			436	517	513	435.5	576	497.5	

Table 2: Bard function ($m = 19$, $\delta = 0.1$) with four added outliers, two above and two below the model curve: first-order method ($B = 0$) versus the second-order variant ($M = 1$). For each o , the “best run” block reports the best $f(x^*)$ over 100 starting points together with the outer iterations (#it) and function evaluations (#fcnt) of the single run that attained it; the remaining columns give the mean and median #it and #fcnt over all 100 starting points. The Σ row sums each count over o .

407 Unlike the polynomial instance, the counts now differ between the two methods, and the
 408 difference has structure. The aggregate over all o is dominated by $o = 0$ —the minimax over
 409 all $m = 19$ observations, outliers included—a hard nonconvex problem where both methods
 410 are expensive (about 360–400 mean iterations) and the first-order method is in fact slightly
 411 cheaper. Once at least the outliers are being discarded ($o \geq 1$), the regime for which the method
 412 is intended, the second-order variant is more efficient in aggregate. The comparison is not

bard_panels.pdf

Figure 1: Bard function with four added outliers, two above and two below the model curve. (a) The fit returned at $o = 4$ (solid line) follows the fifteen clean observations (dots), while the four points identified as outliers (circles) are discarded. (b) Best $f(x^*)$ over the 100 starting points as a function of the number o of discarded observations (logarithmic vertical axis; the two variants coincide): $f(x^*)$ decreases only moderately up to $o = 3$ and drops by more than two orders of magnitude at $o = 4$, after which it stays at the residual floor of the clean data, the onset of which identifies the four outliers.

uniform row by row: the first-order method has the lower mean iteration count at $o = 2$ and $o = 6$, the lower mean evaluation count at $o = 2$, and the lower median iteration count at $o = 5$ and $o = 6$. The one metric that favors the second-order variant at every $o \geq 1$ is the median evaluation count. Summing the per- o means of Table 2 over $o = 1, \dots, 6$ gives 232 iterations and 296 evaluations for the first-order method against 114 and 160 for the second-order variant; the corresponding sums of the per- o medians are 43.5 and 115 against 38 and 83. The mean gap is large—the first-order method uses about twice as many iterations and evaluations—but is amplified by a heavy tail of slow first-order runs on the hardest instances (at $o = 1$ the mean iteration count is 152 for the first-order method against 39 for the second-order variant), which the curvature mitigates. The median, robust to this tail, still favors the second-order variant, though far more modestly. Either way, in aggregate over this regime the curvature lowers the cost.

Curvature thus helps precisely where the first-order model is a poor local approximation—a nonlinear model with outliers still present—while it neither helps on the affine model of Section 4.1 nor on the full-data minimax at $o = 0$. This is consistent with the analysis of Section 3, where the curvature affects only the constants of the complexity bounds: the gain is a constant-factor one, realized here in the outlier-rich regime, not a change in the order. One caveat tempers it: once the outliers are removed ($o \geq 4$) both methods reach the residual floor of the clean data, where the first-order variant attains the lower $f(x^*)$ values, by up to an order of magnitude at $o = 6$. A second caveat, the per-iteration cost of the curvature, is examined next.

4.3 Computational cost

Iteration and evaluation counts do not reflect the extra work the second-order variant performs at every iteration: for each active component it assembles the Hessian $\nabla^2 f_i(x^k)$, diagonalizes it, applies the shift (23) and rescales it, at a cost of one `DSYEV` call per index in $I_\delta(x^k)$. Table 3 therefore reports CPU times, measured with the Fortran intrinsic `cpu_time` around the call to the algorithm and aggregated in the same way as the counts: for each o we take the mean and the median over the 100 starting points, and the table sums those per- o values over the sweep.

	First-order ($B = 0$)		Second-order ($M = 1$)		ratio	
	mean	median	mean	median	mean	median
Polynomial, $o = 0, \dots, 12$	115.7	110.0	158.9	146.3	1.37	1.33
Bard, $o = 0, \dots, 6$	62.2	38.0	51.4	33.8	0.83	0.89
Bard, $o = 1, \dots, 6$	35.1	11.0	29.2	11.7	0.83	1.06

Table 3: CPU time in milliseconds for the polynomial of Section 4.1 and the Bard function of Section 4.2, summed over the sweep in o . For each o the mean and median over the 100 starting points are taken, as in Tables 1 and 2, and those values are then summed. The ratio columns give second-order over first-order, so that values below 1 favour the second-order variant.

The two problems behave in opposite ways, and both are informative. On the polynomial, where the counts were indistinguishable, the second-order variant costs 37% more time in the mean and 33% more in the median: with nothing to gain from the curvature, its assembly and

diagonalization are pure overhead. This quantifies the redundancy diagnosed in Section 4.1. On the Bard problem the iteration savings more than pay for that overhead, and the second-order variant is 17% faster in the mean, both over the whole sweep and over the regime $o \geq 1$. The medians again temper the conclusion: over the whole sweep the second-order variant is 11% faster, but restricted to $o \geq 1$ it is 6% *slower*, because there the typical run of either method terminates in a handful of iterations and the per-iteration overhead is no longer amortized. The wall-clock advantage is therefore real but smaller than the roughly twofold gap in mean iteration counts suggests, and it is confined to the nonlinear problem.

We note that these times are of the order of tens of milliseconds and, being averages over 100 runs of a single sweep, carry some machine noise; the ratios, not the absolute values, are the meaningful quantity.

4.4 Sensitivity to δ and $\sigma_{\min}$

The band half-width δ and the initial regularization $\sigma_{\min}$ are chosen by trial and error, so it is natural to ask how much of the above depends on that choice. We repeated the Bard experiment of Section 4.2 for every combination of $\delta \in \{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$ and $\sigma_{\min} \in \{10^{-1}, 10^{-2}, 10^{-3}\}$, with γ , α , ϵ , M and the 100 starting points unchanged, and with both methods. Table 4 collects the results.

δ	$\sigma_{\min}$	$f(x^*)$ at $o = 3$	$f(x^*)$ at $o = 4$	drop	First-order ($B = 0$)		Second-order ($M = 1$)	
					#it	#fcnt	#it	#fcnt
10^{-1}	10^{-1}	3.511e-01	2.086e-03	$168\times$	232	295	115	160
10^{-1}	10^{-2}	3.536e-01	1.591e-03	$222\times$	54	174	46	130
10^{-1}	10^{-3}	3.493e-01	1.813e-03	$193\times$	36	182	43	158
10^{-2}	10^{-1}	3.491e-01	1.783e-03	$196\times$	258	413	170	285
10^{-2}	10^{-2}	3.489e-01	1.350e-03	$258\times$	83	340	74	281
10^{-2}	10^{-3}	3.489e-01	1.386e-03	$252\times$	65	387	76	340
10^{-3}	10^{-1}	3.487e-01	1.327e-03	$263\times$	332	684	242	523
10^{-3}	10^{-2}	3.488e-01	1.338e-03	$261\times$	148	694	121	551
10^{-3}	10^{-3}	3.488e-01	1.329e-03	$262\times$	110	726	119	639
10^{-4}	10^{-1}	3.487e-01	1.304e-03	$267\times$	427	1158	372	978
10^{-4}	10^{-2}	3.487e-01	1.291e-03	$270\times$	219	1249	178	939
10^{-4}	10^{-3}	3.487e-01	1.293e-03	$270\times$	171	1286	177	1168

Table 4: Sensitivity of the Bard experiment to δ and $\sigma_{\min}$. The values of $f(x^*)$ and the drop factor between $o = 3$ and $o = 4$ are those of the first-order method; the second-order variant agrees with them to within the multistart variability. The count columns give the mean #it and #fcnt over the 100 starting points, summed over $o = 1, \dots, 6$. In all 24 runs the $o = 4$ solution identified exactly the four added outliers.

Three things are stable across the table. The four outliers are identified exactly in every one of the 24 runs. The drop in $f(x^*)$ between $o = 3$ and $o = 4$ is never smaller than two orders of magnitude, ranging from $168\times$ to $270\times$. And the value of $f(x^*)$ at the plateau and at the floor varies only in the third significant digit. The identification of the outliers, which is what

the method is used for, is therefore not an artefact of the particular δ and $\sigma_{\min}$ reported in Section 4.2.

The counts, by contrast, do depend on the parameters, and in a way the analysis anticipates. Decreasing δ raises the evaluation count steeply—from about 300 at $\delta = 10^{-1}$ to about 1200 at $\delta = 10^{-4}$ for the first-order method—which is consistent with the threshold $9c_x^2/(2\delta)$ of Theorem 3.8: a narrower band forces a larger $\sigma_{k,j}$, hence more rejected trial points per iteration. The effect of $\sigma_{\min}$ is the opposite and milder, a smaller initial regularization allowing longer early steps. Notably, the setting $\delta = \sigma_{\min} = 10^{-1}$ used in Section 4.2, inherited from [18], is the one with the largest iteration count in the table; we keep it so that a single parameter set is used throughout the paper and the comparison with the first-order method is made on its own terms.

The comparison between the two methods is likewise stable. The second-order variant has the lower mean evaluation count in *all twelve* parameter combinations, and the lower mean iteration count in eight of the twelve; the four exceptions all occur at the smallest value $\sigma_{\min} = 10^{-3}$, and in each of them the evaluation count still favours the second-order variant. The advantage reported in Section 4.2 is thus not an artefact of the parameter choice either, though its size clearly is.

5 Final remarks

We have extended the first-order regularized method of [18] for box-constrained order-value optimization by enriching its model with curvature, each active component carrying its own positive semidefinite matrix of norm at most M . The resulting method is well defined and, under the same assumptions, attains the same worst-case iteration- and evaluation-complexity bounds as its first-order counterpart (Theorem 3.9), of which it is a strict generalization. The curvature is thus accommodated at no cost in the theoretical guarantees. Positive semidefiniteness and the bound M are the only requirements imposed on the matrices, so quasi-Newton updates such as BFGS or SR1, Gauss–Newton matrices and shifted exact Hessians all fit directly into the framework; only the convex combination of the active matrices enters the analysis, through Corollary 3.6.

The numerical experiments delineate where the curvature helps and where it does not. On the affine model the component Hessians are constant and the curvature is exact but redundant, and the two variants are indistinguishable. On the nonlinear model the picture depends on the regime: for the full-data minimax the curvature does not help, but once outliers are being discarded—the regime for which order-value optimization is intended—the second-order variant is the more efficient of the two, reducing both the mean and the median iteration and evaluation counts aggregated over that regime and, in particular, curbing a heavy tail of slow first-order runs on the hardest instances. The advantage is a constant-factor one, realized where the first-order model is a poor local approximation, and is consistent with the complexity analysis, in which the curvature enters only the constants and not the order of the bounds. It should be read with two caveats. First, the per-iteration cost of assembling, diagonalizing, shifting and rescaling one curvature matrix per active component is not reflected in the evaluation count; measuring it directly (Section 4.3) shows that on the affine model, where the counts are identical, the second-order variant costs some 35% more CPU time, while on the nonlinear model the iteration savings

outweigh the overhead and it is 17% faster in the mean, though only marginally so, or slightly slower, in the median. Second, the solution quality is essentially the same for both variants, with the first-order method reaching somewhat lower values once the outliers have been discarded.

The reason the curvature does not change the order is structural. The Tikhonov term $\sigma\|x - \bar{x}\|^2$ makes the model an upper approximation of f once σ is of order L , which yields a step of size $\|x^{k+1} - x^k\| = O(\epsilon)$ and a per-iteration decrease of order ϵ^2 , hence the $O(\epsilon^{-2})$ iteration bound. A natural way to genuinely improve the order is to replace the quadratic regularization by a cubic one, $\frac{\sigma}{3}\|x - \bar{x}\|^3$, in the spirit of the cubic-regularization method of [16] and the high-order regularized models of [8, 6]. With accurate second-order information—the true or Gauss–Newton Hessian of each active component, used unmodified rather than shifted and rescaled to meet an arbitrary bound M —and under the additional assumption that the component Hessians are Lipschitz continuous, the cubic model is an upper approximation with a step of size $O(\epsilon^{1/2})$ and a per-iteration decrease of order $\epsilon^{3/2}$, which would improve the iteration bound to $O(\epsilon^{-3/2})$. Realizing this for the order-value function is not a direct transfer: the objective is nonsmooth, the complexity carries the two parameters δ and ϵ , and the approximate optimality condition $C_{\delta,\epsilon}$ is stated in terms of active gradients. Extending the analysis to a cubically regularized model, tracking the interaction of the improved ϵ -dependence with δ , is a promising direction in which the curvature would acquire a provable role, at the price of a stronger smoothness assumption, accurate second-order information, and a cubically regularized subproblem.

Declarations

Funding. This work was funded by the Vice-Rectorate for Research, Outreach and Social Projection of the University of the Atlantic through the 2022 Internal Call for Proposals to Strengthen Research Groups with Tools for Creation, Innovation, and Research.

Competing interests. The authors declare that they have no competing interests.

Author contributions. All authors contributed to the conception and design of the study, to the analysis, and to the writing of the manuscript. All authors read and approved the final manuscript.

Data availability. The polynomial data set of Section 4.1 is the one of [1]. The data of Section 4.2 consist of the 15 observations of the Bard function listed in [15] together with four outliers generated with a fixed random seed, as described there. Both data sets are included in the repository cited below.

Code availability. The Fortran 90 implementation of Algorithm 2.1, the data files, and the scripts that reproduce every table and figure of Section 4 are openly available at <https://github.com/gdquintero/ovoqnewton>.

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